Logical Intelligence

Why Logic Matters

Artificial Intelligence

From Search to Reasoning I

In the previous module, we studied search as a mechanism for solving problems.

Search asks:

Which state should I explore next?

Logical reasoning asks a different question:

What follows from what I already know?

From Search to Reasoning II

For example:

All robots require energy	Premise
R2 is a robot	Premise
R2 requires energy	Conclusion

Central idea

Logical intelligence concerns the ability of an agent to derive conclusions that are justified by its knowledge.

Why Do We Need Logical Reasoning? I

An intelligent agent often has to do more than produce a plausible response.
It may need to determine whether a conclusion is actually entailed by its knowledge.

Consider:

$$P \to Q$$
$$Q \to R$$
$$P$$

A reasoning system should be able to derive

$$\boxed{R}$$

because

Why Do We Need Logical Reasoning? II

$$P \Rightarrow Q \Rightarrow R.$$

The important property is not that R is plausible.
It is that

$$\boxed{R \text{ follows necessarily from the premises.}}$$

Question

Can we build AI systems that distinguish between "sounds plausible" and "follows from what is known"?

But What About Large Language Models? I

Modern LLMs can perform remarkably sophisticated reasoning-like tasks.
They can:

- explain concepts;
- solve mathematical problems;
- write programs;
- analyse arguments;
- generate proofs and proof sketches.

So why study logic?

But What About Large Language Models? II

but is it really? We still want to ask:

$\boxed{\text{Does the conclusion actually follow from the premises?}}$

Key distinction

Plausible generation $\neq$ **logical entailment**

A Useful Architecture: Generate, Verify, Revise I

Rather than asking whether we should use

Generative Models or Logic,

we can ask how they might work together.

1. Generate		2. Verify		3. Decide
LLM	$\longrightarrow$	**Logic**	$\longrightarrow$	**Accept / Reject**
Generate a candidate answer		Check whether the claim follows		Is the result sufficiently justified?

If rejected:

A Useful Architecture: Generate, Verify, Revise II

$\boxed{\text{Revise} \quad \longrightarrow \quad \text{Verify again}}$

Key idea

$\boxed{\text{Generate} \rightarrow \text{Verify} \rightarrow \text{Revise} \rightarrow \text{Verify}}$

What Can a Logical Verifier Check? I

Suppose an LLM produces a claim

$$C.$$

If we have a formal knowledge base KB, we can ask whether

$$\boxed{KB \models C}$$

where $\models$ means "entails."

For example:

$$\begin{aligned} KB: \quad & \mathrm{Robot}(R2) \\ & \forall x \left(\mathrm{Robot}(x) \to \mathrm{NeedsEnergy}(x) \right) \end{aligned}$$

The LLM proposes:

What Can a Logical Verifier Check? II

$$C = \mathrm{NeedsEnergy}(R2).$$

We can then determine whether

$$KB \models C.$$

Important

Logical verification does not magically establish that the knowledge base itself describes the real world correctly.

A Simple Example: Detecting a Contradiction I

Suppose our knowledge base contains

$$P \to Q$$

and

$$P.$$

Therefore,

$$Q.$$

Now suppose an LLM generates the statement

$$\boxed{\neg Q}.$$

A Simple Example: Detecting a Contradiction II

The generated answer conflicts with the logical consequences of the knowledge base.

A logical system can identify the inconsistency:

$$KB \models Q$$

while the generated claim is

$$\neg Q.$$

Potential application

Logic can therefore be used as a **verification layer** for certain classes of generated claims.

Where Might Logical Intelligence Be Useful? I

Logical reasoning appears in many different kinds of intelligent systems.

1. **Agents**
   - condition–action rules;
   - knowledge-based agents;
   - reasoning about the environment.
2. **Planning**
   - determining whether actions are possible;
   - reasoning about preconditions and effects.
3. **Verification**
   - checking programs;
   - detecting contradictions;
   - validating generated claims.
4. **Mathematical reasoning**

Where Might Logical Intelligence Be Useful? II

- symbolic algebra;
- theorem proving;
- formal proof assistants.

From Simple Rules to Formal Reasoning I

There is a spectrum of increasingly sophisticated uses of logic.

1 **Simple Reflex** Rules → 2 **Knowledge Based** Agents → 3 **Logical Inference** → 4 **Formal Verification** → 5 **Theorem Proving**

Stage	Example
Simple reflex rules	"If obstacle ahead, turn left"
Knowledge-based agents	Reason about known facts
Logical inference	Derive consequences from premises
Formal verification	Check whether a program satisfies a specification
Theorem proving	Construct and verify mathematical proofs

From Simple Rules to Formal Reasoning II

Key point

The sophistication lies not simply in having rules, but in being able to reason systematically about their consequences.

What Will We Study? I

In this lecture we focus on the use of logic for intelligent reasoning.

Where logical reasoning appears in AI agents:

1. What does a conclusion to follow from what the agent knows?
2. How can an agent use rules to make decisions?
3. How can inference be performed?
4. How do an agent dqpqble of generating statements verify them?

What We Will *Not* Do Yet I

There is an important distinction between

$$\boxed{\text{representing knowledge}}$$

and

$$\boxed{\text{reasoning with knowledge}}.$$

In this lecture we concentrate on the second.

We will not yet focus on:

- learning logical representations;
- constructing large knowledge bases;
- detailed translation from natural language into first-order logic;

What We Will *Not* Do Yet II

- learning rules from data.

Those topics will be introduced later when we study

Logical Representation and Learning.

Today's focus

Given a representation of knowledge, how can an intelligent system use logic to derive, check, and verify conclusions?

The Central Question I

We can now state the central question of this lecture:

How can an AI system reason about what follows from what it knows?

This leads to a progression:

$$\boxed{\text{Rules} \rightarrow \text{Inference} \rightarrow \text{Verification} \rightarrow \text{Formal Reasoning}}$$

For example, in modern AI implementations using LLMs:

The Central Question II

$$\boxed{\text{LLM} \rightarrow \text{Generate} \rightarrow \text{Logic} \rightarrow \text{Verify}}$$

Logic in an Agent I

Recall the basic agent architecture:

$$\boxed{\text{Perceive} \rightarrow \text{Decide} \rightarrow \text{Act}}$$

A simple reflex agent can use condition–action rules:

$$\begin{aligned} \text{ObstacleAhead} &\rightarrow \text{TurnLeft} \\ \text{BatteryLow} &\rightarrow \text{Recharge} \\ \text{Dirty} &\rightarrow \text{Clean} \end{aligned}$$

This is useful, but the agent is essentially reacting to individual conditions.

What if it needs to *combine* several pieces of knowledge?

Logic in an Agent II

Motivation

Logical intelligence allows an agent to derive conclusions rather than merely react to individual percepts.

From Rules to Inference I

Suppose an agent knows:

$$
\begin{aligned}
&P \to Q \\
&Q \to R \\
&P
\end{aligned}
$$

A logical inference mechanism can derive:

$$P \Rightarrow Q \Rightarrow R.$$

Therefore,

$$\boxed{R}$$

is a consequence of what the agent knows.

From Rules to Inference II

The agent did not need a separate rule

$$P \to R.$$

It derived the conclusion by combining existing knowledge.

Key idea

Logical reasoning gives an agent the ability to derive new conclusions from existing knowledge.

Knowledge-Based Agent I

We can therefore separate three components:

$$\boxed{\text{Knowledge} \qquad \text{Inference} \qquad \text{Action}}$$

For example,

$$KB = \left\{ \begin{array}{l} \mathrm{WetRoad} \\ \mathrm{WetRoad} \rightarrow \mathrm{ReducedTraction} \\ \mathrm{ReducedTraction} \rightarrow \mathrm{IncreaseDistance} \end{array} \right\}.$$

The agent can ask:

$$KB \models \mathrm{IncreaseDistance}\ ?$$

The inference mechanism establishes:

Knowledge-Based Agent II

$$\text{WetRoad} \Rightarrow \text{ReducedTraction} \Rightarrow \text{IncreaseDistance}.$$

Thus the agent can decide to increase its following distance.

What Does “Follows From” Mean? I

The central question of logical reasoning is:

$$\boxed{KB \models \alpha\ ?}$$

Read this as:

“Does α follow from the knowledge in KB?”

For example:

$$\begin{aligned} KB: \quad & P \rightarrow Q \\ & P \end{aligned}$$

Then

$$KB \models Q.$$

What Does "Follows From" Mean? II

But if the knowledge base contains only

$$P \rightarrow Q,$$

we cannot conclude Q.

Important

Logical inference is not simply producing a plausible conclusion. It establishes a conclusion from specified premises.

From Logical Agents to Verification I

The same idea applies when the conclusion is not an action, but a claim generated by another AI system.
Suppose an LLM generates

$$C.$$

We can ask a formal reasoning system:

$$KB \models C \ ?$$

The architecture becomes:

$$\boxed{\text{LLM} \rightarrow \text{Candidate Claim} \rightarrow \text{Logical Verification}}$$

For example, if

From Logical Agents to Verification II

$$KB \models Q$$

but the LLM generates

$$\neg Q,$$

the claim is inconsistent with the knowledge base.

Important limitation

This verifies the claim *relative to the knowledge base*. It does not establish that the knowledge base itself is true.

How Do We Perform the Inference? I

We have now reached the central computational question, namely how can an AI system determine whether

$$KB \models \alpha?$$

Three important approaches are:

1. **Forward chaining**
   Start with known facts and derive their consequences.

$$P \to Q, \quad P \quad \Rightarrow \quad Q$$

2. **Backward chaining**
   Start with the desired conclusion and work backwards towards the facts required to establish it.

How Do We Perform the Inference? II

3. **Resolution**
   Transform logical statements and use contradiction to establish whether a conclusion follows.

Next

We now examine these inference mechanisms and what guarantees they provide.

From Entailment to an Inference Procedure I

We have seen that logical reasoning asks whether

$$KB \models \alpha.$$

But an AI system needs a computational procedure for answering this question.

For this lecture, we focus on:

Resolution

From Entailment to an Inference Procedure II

Resolution operates on **clauses**, which are disjunctions of literals.
For example:

$$P \vee Q$$

and

$$\neg Q \vee R.$$

The complementary literals are

$$Q, \qquad \neg Q.$$

The resolution rule is

$$\boxed{(P \vee Q),\ (\neg Q \vee R) \quad \Rightarrow \quad (P \vee R)}$$

Key idea

Resolution provides a mechanical way of deriving logical consequences.

Resolution by Refutation I

Resolution is particularly useful for proving an entailment.
Suppose we want to establish

$$KB \models \alpha.$$

Instead, assume that the conclusion is false:

$$\neg\alpha.$$

We then consider

$$KB \cup \{\neg\alpha\}.$$

If resolution produces a contradiction, then our assumption $\neg\alpha$ must be false.
Therefore:

$$\boxed{KB \models \alpha}$$

whenever

$$\boxed{KB \cup \{\neg\alpha\} \quad \text{leads to a contradiction.}}$$

Resolution by Refutation II

The contradiction is represented by the

$$\boxed{\text{empty clause } \Box}.$$

Strategy

To prove α: assume $\neg\alpha$ and derive $\Box$.

Worked Example: Proving an Entailment I

Suppose

$$KB = \left\{ \begin{array}{l} P \to Q \\ Q \to R \\ P \end{array} \right\}.$$

We want to prove

$$KB \models R.$$

First write the implications as clauses:

$$C_1 : \neg P \lor Q$$
$$C_2 : \neg Q \lor R$$
$$C_3 : P$$

Resolution (contd.) I

$$C_1 : \neg P \vee Q$$
$$C_2 : \neg Q \vee R$$
$$C_3 : P$$

To prove R, add $C_4 : \neg R$
Now apply resolution:

$$C_2, C_4 \Rightarrow \neg Q$$
$$C_1, \neg Q \Rightarrow \neg P$$
$$C_3, \neg P \Rightarrow \square$$

We have derived the empty clause.
Therefore,

Resolution (contd.) II

$$\boxed{KB \models R.}$$

What Just Happened? I

The resolution proof can be viewed as:

$$\frac{\begin{array}{c} P \rightarrow Q \\ Q \rightarrow R \\ P \end{array}}{\text{Assume } \neg R}$$

$$\downarrow$$

$$\neg Q$$

$$\downarrow$$

$$\neg P$$

$$\downarrow$$

$$P \wedge \neg P$$

$$\downarrow$$

$$\square$$

What Just Happened? II

The contradiction tells us that $\neg R$ cannot be true if the original knowledge base is true

Theorem-Proving I

We just showed

$$\boxed{R \text{ follows from } KB.}$$

That is, R is a *theorem* derivable from the axioms in KB This illustrates an important idea:

$$\boxed{\text{Logical entailment} \quad \longrightarrow \quad \text{mechanical inference}}$$

A computer does not need to "understand" the argument in the human sense. It can apply a formal inference procedure.

Why Resolution Matters for Modern AI I

Resolution provides a bridge between

logic

and

computation.

Resolution can be used to verify/refute logical claims:

$$\boxed{\text{LLM} \rightarrow \text{Candidate Claim} \rightarrow \text{Resolution} \rightarrow \text{Verified / Rejected}}$$